

The SIMC Framework

**Aligning Academic Research and
Government Practice in Field Experiments**

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Prepared for:

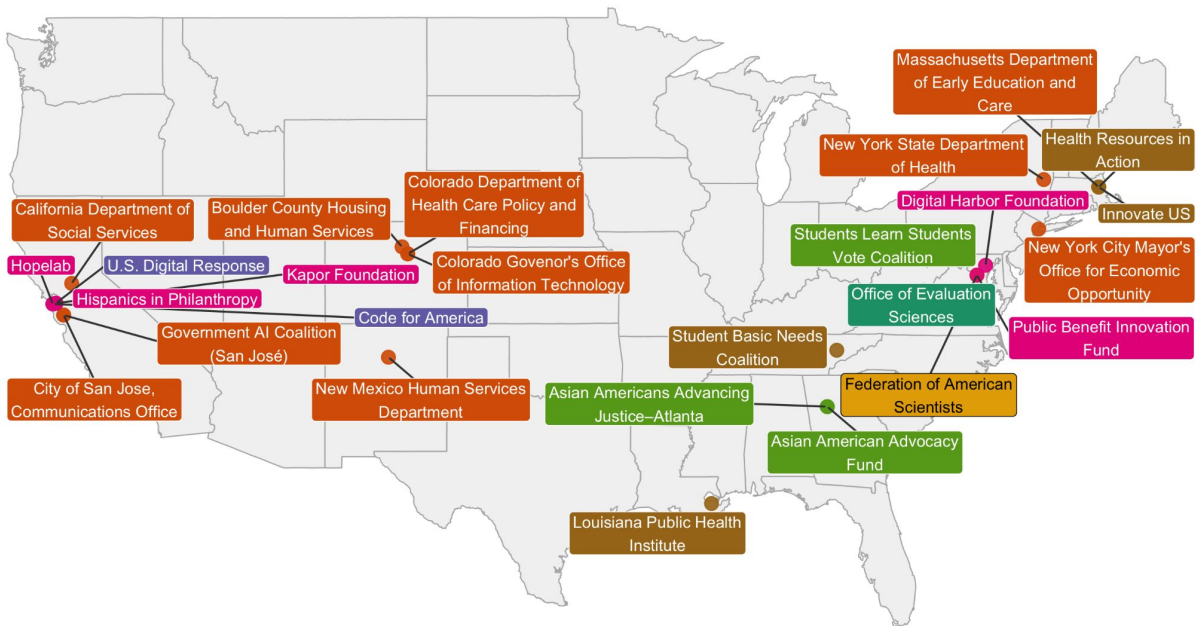
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Acknowledgement



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Category

● Federal government	● State & local government	● Civic tech	● Nonprofits
● Philanthropy	● Advocacy organizations	● Think tanks	

- 1. Field experiments are effective learning and communication tools for learning (evaluating) and doing (implementing) policy.**
- 2. Field experiments are hard, not in the way theoretical physics is hard, but in the way marriage and parenting are hard.**
- 3. The key to successful field experiments is “alignment.”**
- 4. Alignment is a continuous process.**
- 5. Alignment is a mindset, framework, and also practice.**

Field experiments in governments

"I think when people think about Silicon Valley they imagine Steve Jobs in a garage, or the invention of the iPhone or the iPad, and they think that's what Silicon Valley is. ... Most of the innovation in big tech companies is incremental. **The A/B test** is probably **the most impactful business process innovation** in a very long time."

- Susan Athey (Stanford Business School, previously Chief Economist at Microsoft and the U.S. Department of Justice Antitrust Division)

- 1. Field experiments are informative because the evidence comes from the real world.**
- 2. Field experiments are credible because the assumptions are relatively straightforward.**
- 3. Field experiments are easy to communicate and thus defend.**
- 4. Modern technology and data infrastructure make experiments easier and cheaper to run, scale, and adapt.**
- 5. And yet, as I will show you, they can still be *difficult* to do.**

**Instead of using frames such as
“*evidence-based*” or “*data-driven*,” I will
emphasize how field experiments are linked
to “*learning by doing*” in policymaking.**

The Science of “Muddling Through”

By CHARLES E. LINDBLOM

*Associate Professor of Economics
Yale University*

SUPPOSE an administrator is given responsibility for formulating policy with respect to inflation. He might start by trying to list all related values in order of importance, e.g., full employment, reasonable business profit, protection of small savings, prevention of a stock market crash. Then all possible policy outcomes could be rated as more or less efficient in attaining a maximum of these values. This would of course require a prodigious inquiry into values held by members of society and an equally prodigious set of calculations on how much of each value is equal to how much of each other value. He could then proceed to outline all possible policy alternatives. In a third step, he would

➤ Short courses, books, and articles exhort administrators to make decisions more methodically, but there has been little analysis of the decision-making process now used by public administrators. The usual process is investigated here—and generally defended against proposals for more “scientific” methods.

Decisions of individual administrators, of course, must be integrated with decisions of others to form the mosaic of public policy. This integration of individual decisions has become the major concern of organization theory, and the way individuals make decisions necessarily affects the way those decisions are best meshed with others'. In addition, decision-making method relates to allocation of decision-making responsibility—who should make what decision.

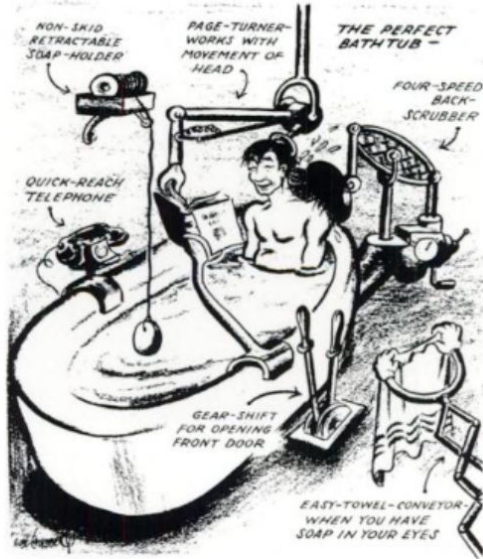
More “scientific” decision-making also is discussed in this issue: “Tools for Decision-Making in Resources Planning.”

How should
policy makers
make decisions
under
uncertainty?

Local search,
iterative
adaptation,
Collective
intelligence

IMPLEMENTATION

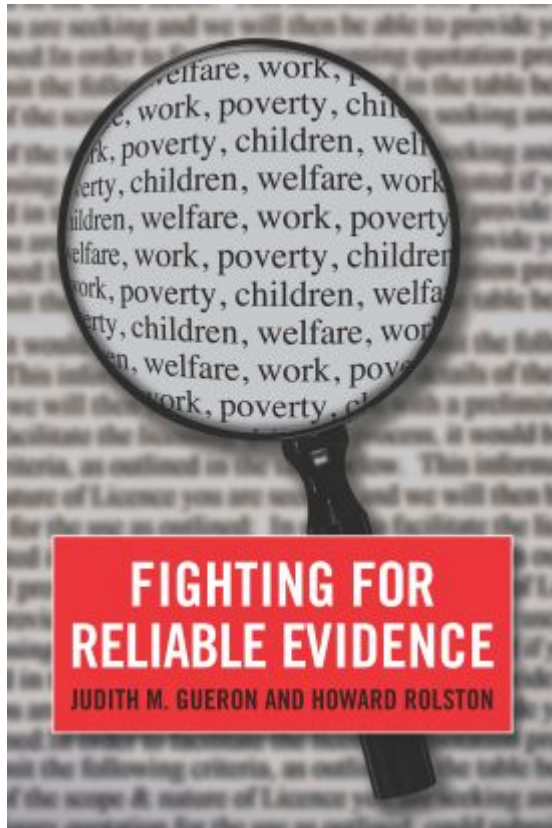
Jeffrey L. Pressman & Aaron Wildavsky



How Great Expectations in Washington
Are Dashed in Oakland; Or,
Why It's Amazing that
Federal Programs Work at All,
This Being a Saga of the
Economic Development Administration
as Told by Two Sympathetic Observers
Who Seek to Build Morals on a
Foundation of Ruined Hopes

Evaluation (*learning*)
and implementation
(*doing*) are two sides
of the same coin:
polycraft.

The history of using field experiments to evaluate evidence in policymaking goes back to the 1960s, around the same time that policy programs began to emerge.



- **Background: 40 decades of program evaluation**
- **Co-authors: Gueron (MDRC) and Rolston (Department of Human and Health Services)**

RESEARCH ARTICLE

Administrative checkpoints, burdens, and human-centered design: Increasing interview access to raise SNAP participation

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Abstract

In this study we describe the potential of human-centered design principles to identify burdens, reducing the effects of administrative checkpoints. Administrative checkpoints—mandatory requirements that must be satisfied in order to progress in an administrative process—have disproportionate negative effects in excluding the public from receiving public services. Mandatory interviews are one such checkpoint. Based on consultation with safety net clients and caseworkers, we designed a field experiment ($N = 1,554$) to minimize the exclusionary effects of mandatory interviews for SNAP applicants. Compared to a control group that received a traditional mailer reminder, SNAP applicants who also received texts reminding them of the interview and communicating flexible “interview anytime” scheduling options had a higher interview completion rate by 10 percentage points, a higher benefit-approval rate by 6 to 7 percentage points, and also completed interviews 3 to 4 days sooner. Follow-up surveys show that the text reminders reduced learning costs about the interview requirement and increased applicant confidence about their ability to manage the process.

Since the 2010s, civic tech and human-centered design have expanded the focus from “*Does the program work?*” to “*Can we improve how it works?*”

Example:
Supplemental Nutrition Assistance Program



North Carolina Department of Health and Human Services Application for Food and Nutrition Services

Do You Need An Interpreter To Help You Apply For Food and Nutrition Services?

Please tell us if you need assistance because you do not speak English or have a disability. Free language assistance and/or other aids and services are available upon request. To receive free interpreter services, call 866-719-0141 or call your local DSS office at _____. After the recorded message, you will reach an operator who can provide you with an interpreter. If you have a disability and need communication assistance, call 866-719-0141 or Relay Services: 711.

What are Food and Nutrition Services?

Food and Nutrition Services help households buy eligible food in authorized retail food stores. This will increase low-income household's food buying power, so they can have more nutritious meals.

How Do I Apply for Food and Nutrition Services?

Step 1. Fill out this application: You have the right to receive an application upon request. If you cannot complete this application, you will only need to provide a signature, legible name, and address. If you need assistance in completing this form, please let us know so that we can assist you.

Step 2. Turn in the application to your local agency as soon as possible: You can mail, fax or bring the application to your local agency or apply online at <https://epass.nc.gov>. The date we get your application with your name, address, and signature on it is also the start date of your Food and Nutrition Services application. If you are eligible for Food and Nutrition Services in the month you apply, the amount of Food and Nutrition Services you will get for that month depends on the date you turn in your application. The sooner you give us this application, the quicker you will know if you are able to get Food and Nutrition Services.

Step 3. Talk with us: A caseworker must interview you or someone you choose to represent you. This is to see if you can get Food and Nutrition Services. If you are unable to stay for your interview today, please tell the receptionist or caseworker so that we can schedule an interview for you.

Information About Social Security Numbers, US Citizenship and Immigration Status

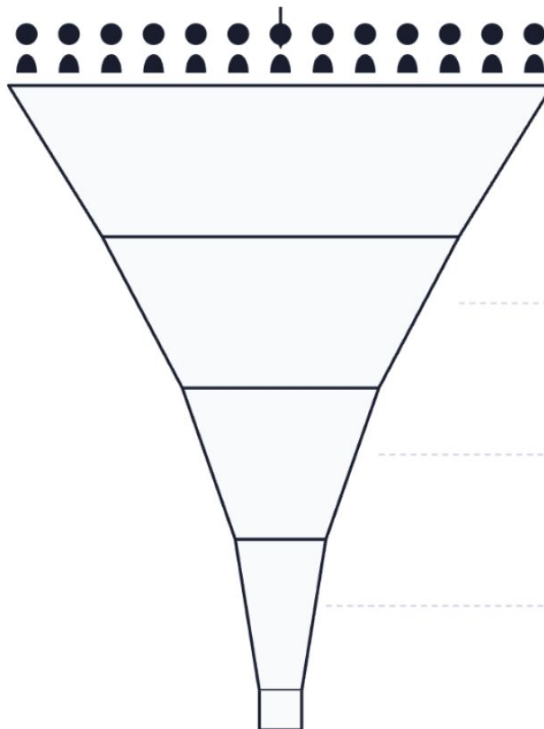
For everyone that you are applying for, you must provide information about Social Security Numbers (SSNs) and citizenship/immigration status. If you do not want to answer questions about SSNs or citizenship/immigration status, you may choose not to apply. Providing an SSN is required by the Food and Nutrition Act for applicants seeking benefits. We will not share SSNs with US Citizenship and Immigration Service (USCIS). We will only use the SSNs you give us to do computer matches to check what you told us with State and Federal Agencies, Income and Eligibility Verification System (IEVS), other computer matching systems, program reviews and audits. This information may be verified through other sources when discrepancies are found and may affect your household's eligibility and benefit level. You must be a United States (U.S.) citizen or an eligible alien and also meet other Food and Nutrition Services rules to get Food and Nutrition Services benefits. We will only contact USCIS to check the immigration status on the household members who give us their immigration documents. If an applicant does not provide this information, they will be ineligible for benefits. Household members must provide their financial information because it is needed to determine eligibility for individuals who are applying. Eligible household members who apply will be able to get benefits even though some people in the household are not applying for benefits. The amount of benefits will depend on the number of people requesting benefits.

Tell Us Do you need someone to apply for or use your Food and Nutrition Services?

If you want someone other than yourself to apply for, use, or obtain information about your benefits, please check yes below. If you check **Yes**, we will give or mail you a form. You and the person you want to help can complete the form and return it to our office. If you choose, this person will receive an EBT card and will have access to your Food and Nutrition Services. An Electronic Benefit Transfer Card (EBT) is a plastic card you use at the store to buy food. Do you need an Authorized Representative to help you get and/or use your Food and Nutrition Services? ☐ Yes ☐ No

APPLY

Online in **48 of 50** states



Application

Up to 1 hour to complete

Mandatory interview

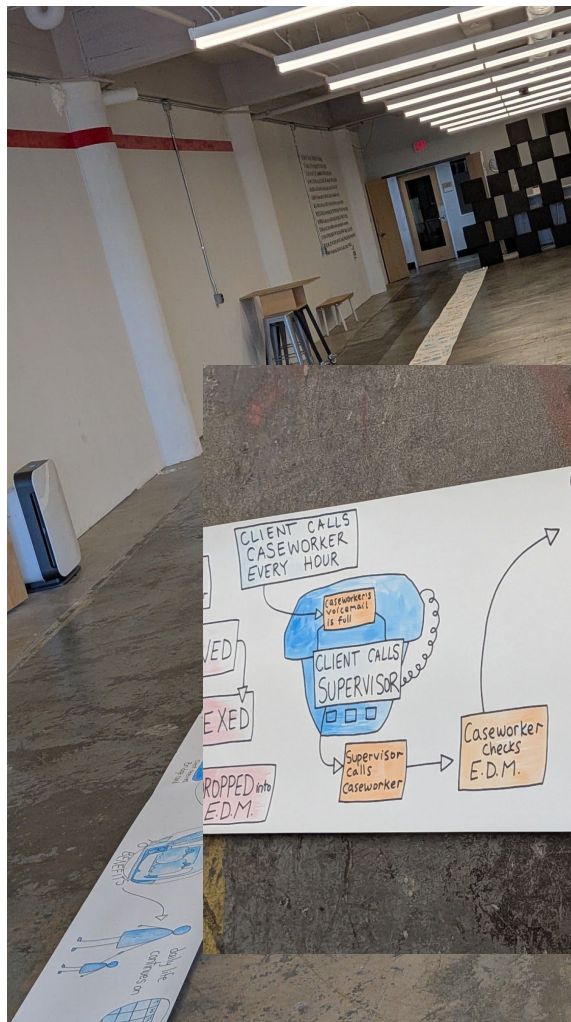
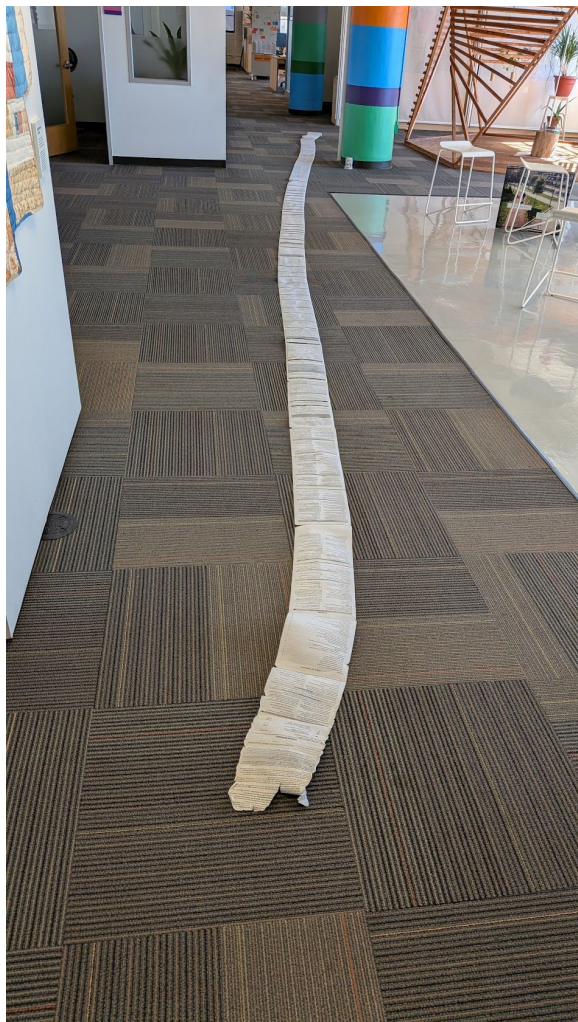
Must be completed within 30 days

Document verification

Pay stubs, ID, lease, utility bills

Renewal

Every 6-12 months



All photos by me,
taken at Civilla, a
Detroit-based design
studio in 2026.

Why field experiments is hard

“In our view, the most important group—the main heroes in this book—were ***the state and local administrators*** who displayed ***the courage, determination, and political skill*** needed to put their programs on the line, and in some cases, win the battles that erupted when we sought to open up real-world programs to a new type of scrutiny.” - *Fighting for Reliable Evidence*, Gueron and Rolston (2013: 428)

The missing semester in field experimentation: Relationship building

“The scientists then in the [CMU] Engineering College neither understood engineering nor believed it could be taught. ***They educated engineers by giving them a lot of physics and math, hoping that their students would later be able to design safe bridges or airplanes.***” - *Models of My Life*, Herbert A. Simon (1996: 258)

If there's one takeaway, it's this:
Alignment

PASTELID'S QUA

*Bas
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In*

RESEARCHER
PERSPECTIVE

Quest for
fundamental
understanding?

YES

NO

PRACTITIONER PERSPECTIVE

Consideration of practical use?

NO

YES

FUNDAMENTAL UNDERSTANDING: YES

Pure basic research

Advances knowledge without immediate practical use.

FUNDAMENTAL UNDERSTANDING: YES

Use-inspired basic research

Builds knowledge while addressing a practical need.

FUNDAMENTAL UNDERSTANDING: NO

No defined category

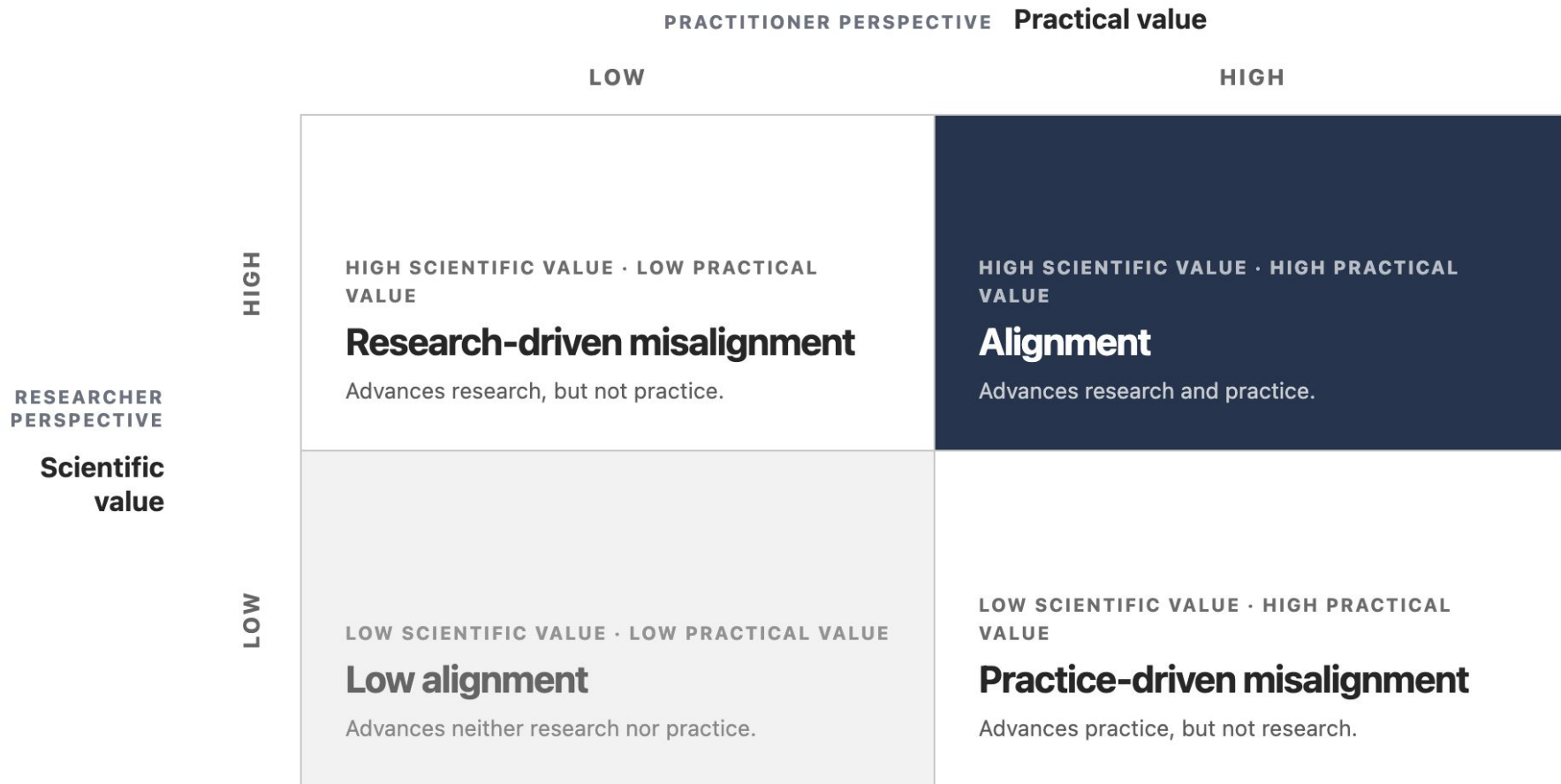
Neither fundamental nor directed toward practical use.

FUNDAMENTAL UNDERSTANDING: NO

Pure applied research

Solves a practical problem using existing knowledge.

Do



If there's another takeaway, it's this:
Continuous *alignment*

SIMC (“sync”) framework

A PRACTICAL FIELD GUIDE

Co-Producing Field Experiments

Using the SIMC framework to align scientific rigor with practical value.

Collaboration works when researchers and practitioners have the **capability, opportunity, and motivation** to do the work together.

SIMC

01

Scoping

Align scientific and practical value

02

Implementation

Balance rigor and feasibility

03

Measurement

Define credible, useful outcomes

04

Communication

Turn learning into action

Start with a problem worth solving together

Co-define the problem, intervention options, goals, and constraints before the design hardens.

WHAT MISALIGNMENT LOOKS LIKE

Research-driven

The design is rigorous, but too disruptive, risky, or costly.

Practice-driven

The fix is feasible, but produces little new knowledge.

HOW TO BUILD ALIGNMENT

1 Build a shared vocabulary. Turn practical problems into testable questions.

2 Offer a portfolio. Compare feasibility and scientific value.

3 Learn the environment. Study workflows and frontline constraints.

4 Revisit priorities. Keep negotiation active as the project evolves.

Capability

Express shared needs

Opportunity

Meet before design

Motivation

Surface disagreement

Offer a burden-reduction portfolio

Different interventions trade implementation demand for depth and durability of impact.

INTERVENTION	TYPICAL PROBLEM	DEMAND	POTENTIAL IMPACT
01 Messaging Reminders and benefit framing	Applicants miss key steps or deadlines.	Low	Small–moderate
02 Redesign forms & flows Simplified forms and verification	Application steps are too complex.	Moderate	Mixed
03 Automate workflows Automatic renewals and data matching	Renewals create recurring burdens.	High	Large

Design around operations—not ideal conditions

Credibility depends on consistent delivery, accurate data, and designs that fit legal and operational realities.

TWO FAILURE MODES

Research-driven

Strict protocols conflict with rules, duties, or data systems.

Practice-driven

Early expansion or missing support weakens fidelity.

A PRACTICAL IMPLEMENTATION KIT

BRIEF

Document the work

Summarize logic, delivery, timing, and staffing in 1–2 pages.

TEAM

Bring in the full system

Engage leaders, frontline staff, IT, legal, and data teams.

PILOT

Test before scaling

Use pilots and shared logs to find failures early.

ADAPT

Preserve the contrast

When pure control is impossible, test an added encouragement.

Capability

Understand workflows

Opportunity

Fit regular operations

Motivation

Respond visibly

Make outcomes credible and useful

Jointly define outcomes that can support causal claims and answer the agency's real decisions.

WHAT PULLS TEAMS APART

Research-driven

Rich outcomes require data the agency cannot support.

Practice-driven

Easy metrics replace meaningful behavioral outcomes.

BUILD A SHARED MEASUREMENT STACK

PRIMARY

Trusted behavioral outcome

Choose a valued outcome the system captures reliably.

SECONDARY

Mechanisms and added learning

Add focused surveys to explain mechanisms.

FOUNDATION

Access, definitions, and a PAP

Verify fields and agree on outcomes before launch.

Capability

Interpret data honestly

Opportunity

Secure early access

Motivation

Own learning together

Share learning in forms people can use

Communication begins before results arrive and continues long before an academic article is published.

REPLACE THIS

Academic default

Long reports

Statistical jargon

Binary success claims

Publication timelines



WITH THIS

Decision-useful communication

One-page briefs and reusable slides

Plain language and one clear figure

Null results framed as learning

Timely updates

Use language that lowers risk:

Pilot

Learning test

Proof of concept

Capability

Translate accurately

Opportunity

Agree on format and timing

Motivation

Share mixed findings

Alignment is built at every stage

Use this checklist to diagnose what the partnership needs next.

STAGE	CAPABILITY	OPPORTUNITY	MOTIVATION
Scoping	Shared language	Early, recurring meetings	Surface disagreement
Implementation	Workflow knowledge	Fit within operations	Respond to constraints
Measurement	Data literacy	Access before launch	Own outcomes together
Communication	Plain-language translation	Practitioner-ready formats	Value learning, not headlines

Partnership is the foundation for
effective, collective, and iterative
learning by doing.

Keep *learning.*

Keep *doing.*

Keep *building stronger partnerships.*

Send me comments / questions (7-11-5):
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